Engineering Multiagent Systems: an Introduction



Viviana Mascardi, University of Genova, Italy

EASSS 2024, August 22nd

Intelligent Agents and Multiagent Systems

N. Jennings, K. Sycara, M. Wooldridge, JAAMAS 1(1), 1998

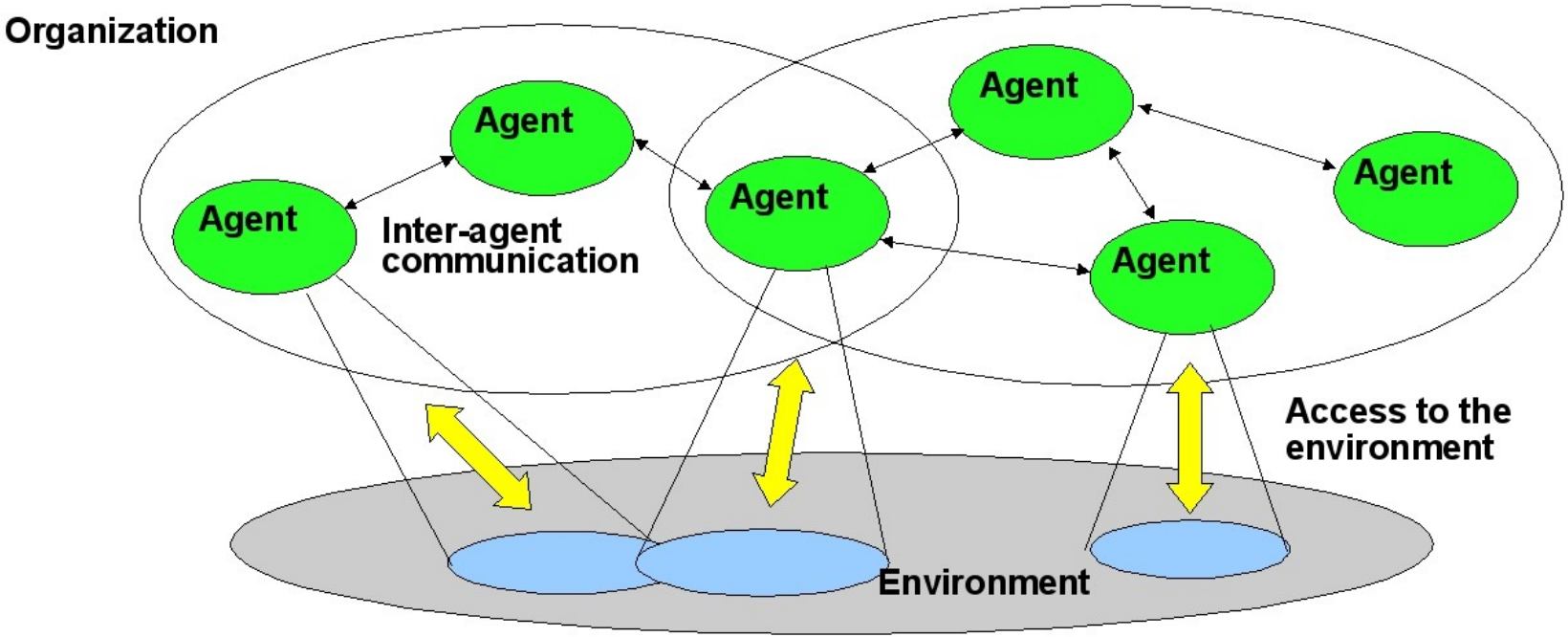
An **agent** is a hardware or software system

- situated
- autonomous
- flexible
  - reactive
  - proactive
  - social

The **strong definition** of agency assumes that agents are conceptualized using mentalistic notions usually attributed to humans (beliefs, desires, intentions, goals, plans).

A **multiagent system (MAS)** is a distributed system where

- each agent has incomplete information, or capabilities for solving the problem, thus each agent has a limited viewpoint;
- there is no global system control;
- data is decentralized; and
- computation is asynchronous.



Zambonelli, F., Omicini, A. Challenges and Research Directions in Agent-Oriented Software Engineering. Autonomous Agents and Multi-Agent Systems 9, 253–283 (2004).

Plus…

People and organizations (**EASSS 2024, Monday**)

Reinforcement Learning (RL), Deep RL, MARL, Continual Learning (**EASSS 2024, Tuesday**)

Coordination, Organisations, Institutions, Norms and Ethics (**EASSS 2024, Wednesday**)

Argumentation (**EASSS 2024, Wednesday**)

Agent-based modeling and simulation (**EASSS 2024, tomorrow**)

Our fictional scenario

Dr. Joyce, the new major of SteaMas, wants to make SteaMas green, smart, safe by ensuring that

1. SteaMas citizens and tourists respect well defined civil/ethical norms

2. no private traffic exists: citizens and tourists may take advantage of different kinds of free public transportation means in the most effective way

3. both citizens and tourists may be driven around by a personal drone provided by the SteaMas municipality for free

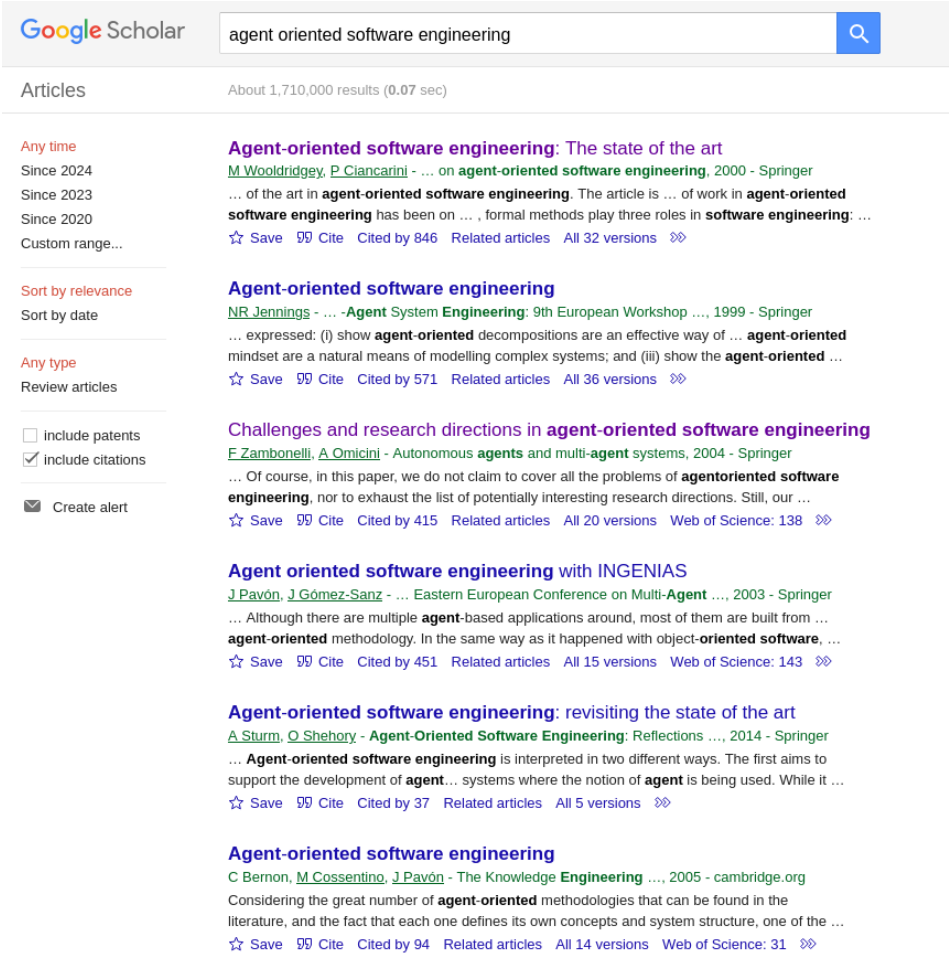
Dr. Joyce is aware of the existence of agents and multiagent systems. He believes that the MAS technology can suit his needs and contacts us to exploit this technology for SteaMas



Back to EMAS literature

Some relevant research directions in Agent-Oriented Software Engineering (AOSE)

1) Agent-oriented architectures
2) AOSE methodologies and notation techniques
3) Agent-oriented programming languages
4) MAS infrastructures and toolkits
5) Agent-based modeling and simulation
6) Integration of symbolic and sub-symbolic approaches for agents
7) Engineering safe autonomous systems


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Agent-oriented software engineering: The state of the art
M Wooldridgey, P Ciancarini - … on agent-oriented software engineering, 2000 - Springer
… of the art in agent-oriented software engineering. The article is … of work in agent-oriented software engineering has been on … , formal methods play three roles in software engineering: …
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Agent-oriented software engineering
NR Jennings - … -Agent System Engineering: 9th European Workshop …, 1999 - Springer
… expressed: (i) show agent-oriented decompositions are an effective way of … agent-oriented mindset are a natural means of modelling complex systems; and (iii) show the agent-oriented …
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Challenges and research directions in agent-oriented software engineering
F Zambonelli, A Omicini - Autonomous agents and multi-agent systems, 2004 - Springer
… Of course, in this paper, we do not claim to cover all the problems of agentoriented software engineering, nor to exhaust the list of potentially interesting research directions. Still, our …
Save Cite Cited by 415 Related articles All 20 versions Web of Science: 138
Agent oriented software engineering with INGENIAS
J Pavón, J Gómez-Sanz - … Eastern European Conference on Multi-Agent …, 2003 - Springer
… Although there are multiple agent-based applications around, most of them are built from … agent-oriented methodology. In the same way as it happened with object-oriented software, …
Save Cite Cited by 451 Related articles All 15 versions Web of Science: 143
Agent-oriented software engineering: revisiting the state of the art
A Sturm, O Shehory - Agent-Oriented Software Engineering: Reflections …, 2014 - Springer
… Agent-oriented software engineering is interpreted in two different ways. The first aims to support the development of agent… systems where the notion of agent is being used. While it …
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Agent-oriented software engineering
C Bernon, M Cossentino, J Pavón - The Knowledge Engineering …, 2005 - cambridge.org
Considering the great number of agent-oriented methodologies that can be found in the literature, and the fact that each one defines its own concepts and system structure, one of the …
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[BOOK] Multi-agent oriented programming: programming multi-agent systems using JaCaMo
O Boissier, RH Bordini, J Hubner, A Ricci - 2020 - books.google.com
… , intelligent, goal-oriented entities called agents, communicating with each other and … of multi-agent oriented programming,(MAOP) which supports the multi-agent systems paradigm at …
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A review of agent-based programming for multi-agent systems
RC Cardoso, A Ferrando - Computers, 2021 - mdpi.com
… A survey on agent-oriented programming from the software engineering perspective can be … to identify the fundamental features of agent oriented programming, and more importantly, …
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A review of platforms for the development of agent systems
CV Pal, F Leon, M Paprzycki, M Ganzha - arXiv preprint arXiv:2007.08961, 2020 - arxiv.org
… agents as a metaphor and implement agent systems using any programming environment, usually an object-oriented … , in dealing with the specific aspects of agent systems. On the other …
Save Cite Cited by 52 Related articles All 6 versions
[HTML] Intelligent software engineering in the context of agile software development: A systematic literature review
M Perkusich, LC e Silva, A Costa, F Ramos… - … and Software …, 2020 - Elsevier
… First, as the application of artificial intelligence (AI) technologies to software engineering. Second, as the development of software engineering solutions for intelligent software. In this …
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Multi-agent systems and complex networks: Review and applications in systems engineering
M Herrera, M Pérez-Hernández, A Kumar Parlikad… - Processes, 2020 - mdpi.com
… This paper presents a review of how multi-agent systems and complex networks theory are brought together to address systems engineering and management problems. The review …
Save Cite Cited by 110 Related articles All 13 versions Web of Science: 54
Logic-based technologies for multi-agent systems: a systematic literature review
R Calegari, G Ciatto, V Mascardi, A Omicini - … Agents and Multi-Agent …, 2021 - Springer
… Also, the long-term research activity in fields like agent-oriented … agent-oriented software

[+] [–] International Workshop on Engineering Multi-Agent Systems (EMAS)

[–] Venue Information

- *predecessor:* International Workshop on Agent-Oriented Software Engineering (AOSE)
- *predecessor:* International Workshop on Declarative Agent Languages and Technologies (DALT)
- *predecessor:* International Workshop on Programming Multi-Agent Systems (ProMAS)

11th EMAS 2023: London, UK

Andrei Ciortea, Mehdi Dastani, Jieting Luo:
Engineering Multi-Agent Systems - 11th International Workshop, EMAS 2023, London, UK, May 29-30, 2023, Revised Selected Papers. Lecture Notes in Computer Science 14378, Springer 2023, ISBN 978-3-031-48538-1 [contents]

9th EMAS@AAMAS 2021: Virtual Event

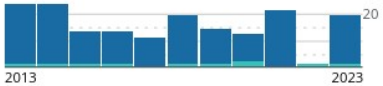
Natasha Alechina, Matteo Baldoni, Brian Logan:
Engineering Multi-Agent Systems - 9th International Workshop, EMAS 2021, Virtual Event, May 3-4, 2021, Revised Selected Papers. Lecture Notes in Computer Science 13190, Springer 2022, ISBN 978-3-030-97456-5 [contents]

8th EMAS@AAMAS 2020: Auckland, New Zealand

Cristina Baroglio, Jomi Fred Hübner, Michael Winikoff:
Engineering Multi-Agent Systems - 8th International Workshop, EMAS 2020, Auckland, New Zealand, May 8-9, 2020, Revised Selected Papers. Lecture Notes in Computer Science 12589, Springer 2020, ISBN 978-3-030-66533-3 [contents]

[–] Venue statistics

records by year



document access

- closed (158)
- open (1)

frequent authors

- Jomi Fred Hübner (13)
- Alessandro Ricci (12)
- Rafael H. Bordini (10)
- Matteo Baldoni (8)
- Mehdi Dastani (8)
- Amal El Fallah Seghrouchni (8)
- Brian Logan 0001 (7)
- Louise A. Dennis (7)
- Michael Winikoff (7)
- Viviana Mascardi (6)
- Cristina Baroglio (6)
- Angelo Ferrando 0001 (6)
- Rafael C. Cardoso 0001 (6)
- Koen V. Hindriks (5)
- Jaime Simão Sichman (5)
- Jürgen Dix (5)
- Michael Fisher 0001 (5)

Engineering and Analysis of Multiagent Systems (EMAS)

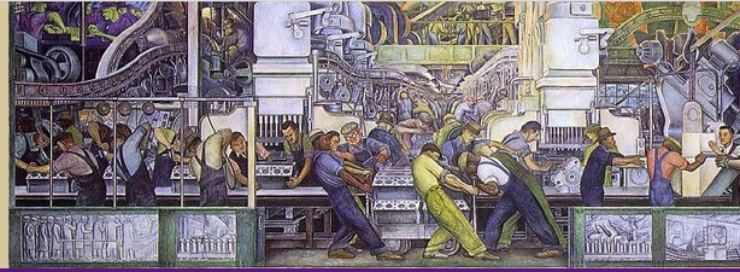
Area Chairs: Viviana Mascardi, Daniela Briola



Topics:

- Requirements and formal specification
- Architecture and modeling
- Formal verification and validation
- Programming models and languages
- Testing, maintenance, and evolution
- Concurrency, fault tolerance, robustness, reliability, performance, and scalability
- Sociotechnical systems, norms, and governance
- Responsibility and accountability
- Interoperability, business agreements, and interaction protocols
- Declarative, Logic-based, and BDI-based agents
- Engineering ethical agents
- Engineering MAS-based simulations
- Tools and testbeds
- Technological paradigms, including microservices, the Web, the IoT, Cloud computing, distributed Ledgers, and Robotics
- Middleware and platforms for MAS
- Engineering learning agents
- Usability
- Applications, including Finance, Health, Agriculture, Autonomous Vehicles and Smart-*

DETROIT aamas2025



24th International Conference on Autonomous Agents and Multiagent Systems

Call for Papers (Main Technical Track)

All submissions will be rigorously peer-reviewed and evaluated on the basis of the overall quality of their technical contribution, taking into account criteria such as originality, significance, soundness, reproducibility, clarity, relevance to the conference, quality of presentation, as well as understanding and appropriate referencing of the state of the art.

The papers will be published under CC BY license.

Important Dates

- **Abstract submission:** Oct 9, 2024
- **Paper submission:** Oct 16, 2024
- **Rebuttal period:** Nov 27 – Dec 4, 2024
- **Author notification:** Dec 23, 2024
- **Camera-ready paper:** Feb 7, 2025



SEARCH

NEWS & UPDATES

Registration opens Fall 2024

IMPORTANT DATES

AAMAS Conference Dates
May 19–23, 2025

May 19–20: Tutorials & Workshops

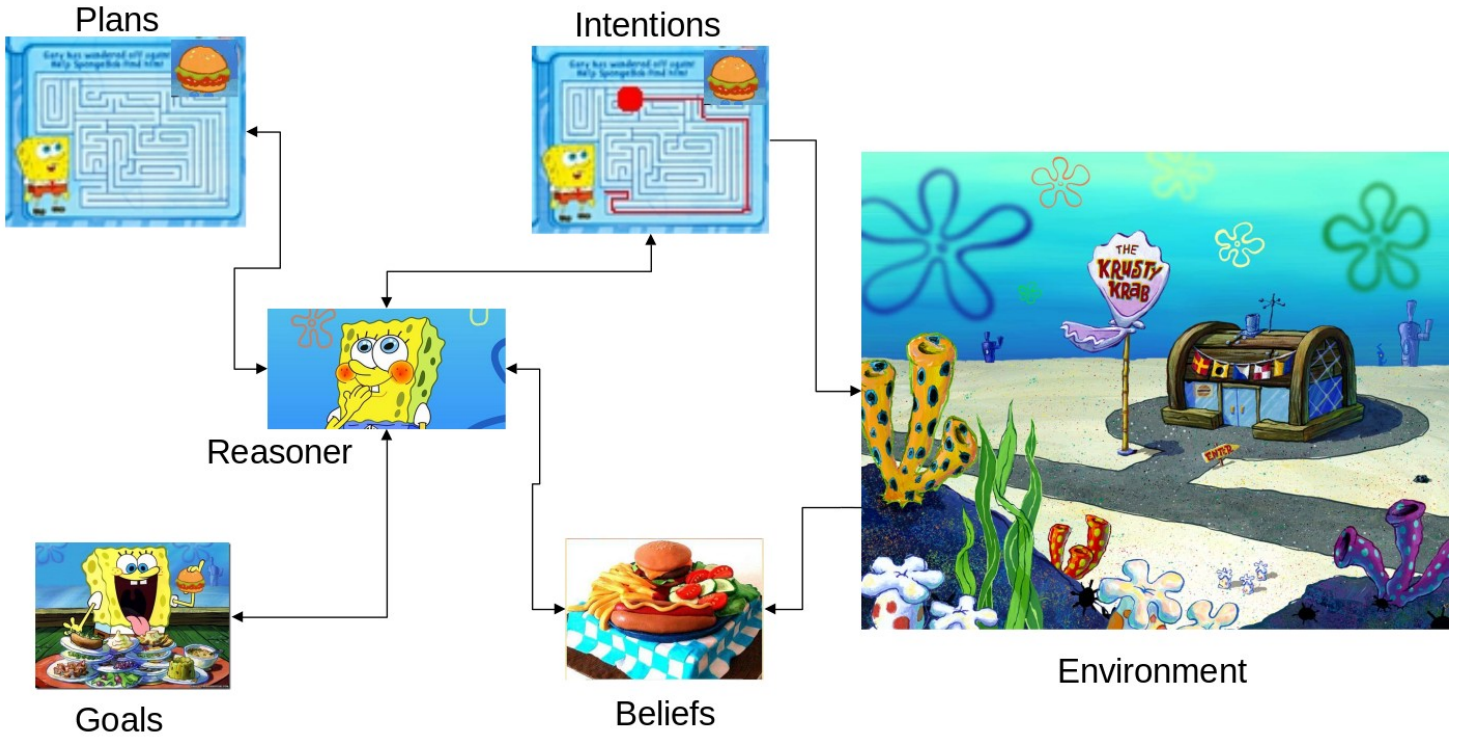
May 21–23: Main Conference

Submission Deadlines

Agent-oriented architectures

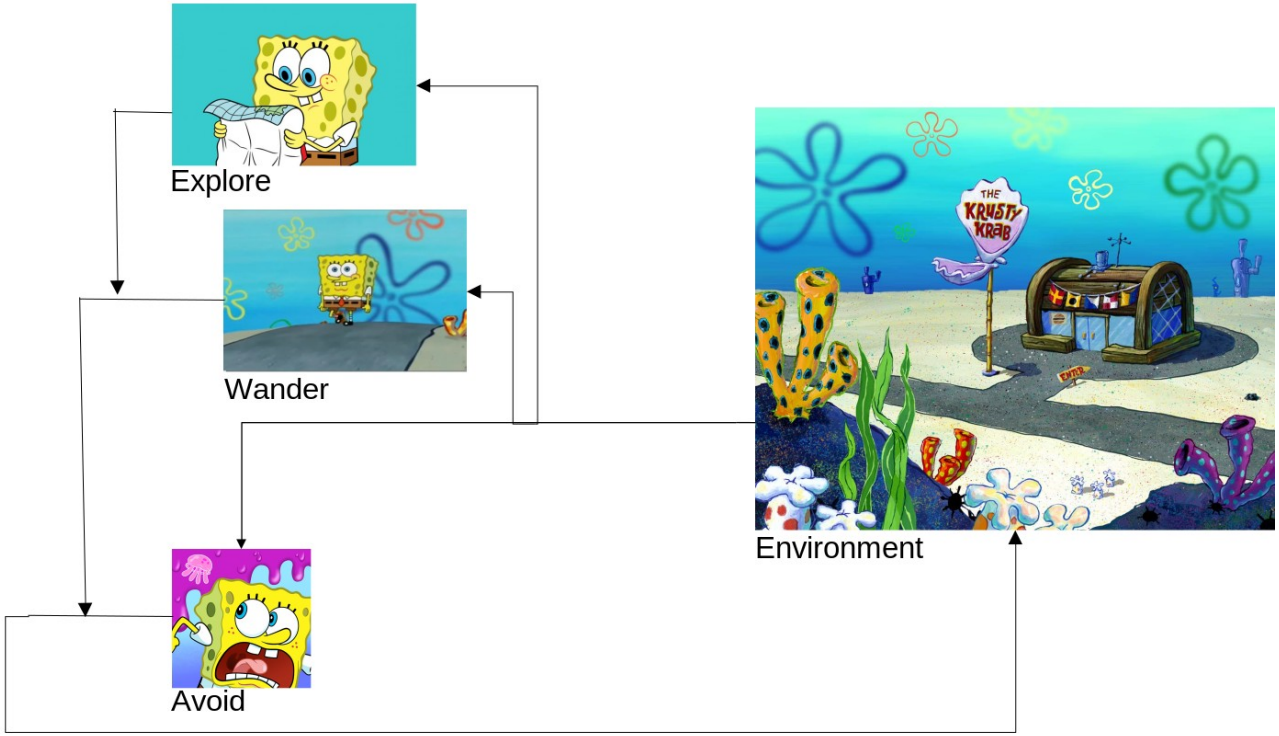
Beliefs-Desire-Intention architecture

A. S. Rao, M. P. Georgeff. (1995)
"BDI-agents: From Theory to Practice", ICMAS'95 (+ Georgeff & Lansky 1986, Rao and Georgeff 1992, Shoham 1993, ...)



Subsumption architecture

Brooks, R. (1986). "A robust layered control system for a mobile robot". IEEE Journal of Robotics and Automation. 2 (1): 14–23



Hybrid architectures

I. A. Ferguson, "Touring Machines: autonomous agents with attitudes," in Computer, vol. 25, no. 5, pp. 51-55, (1992)

Müller, J. and Markus Pischel. “The agent architecture InteRRaP: concept and application.” (1993).

Hybrid architectures

I. A. Ferguson, "Touring Machines: autonomous agents with attitudes," in Computer, vol. 25, no. 5, pp. 51-55, (1992)
Müller, J. and Markus Pischel. “The agent architecture InteRRaP: concept and application.” (1993).



Hybrid architectures

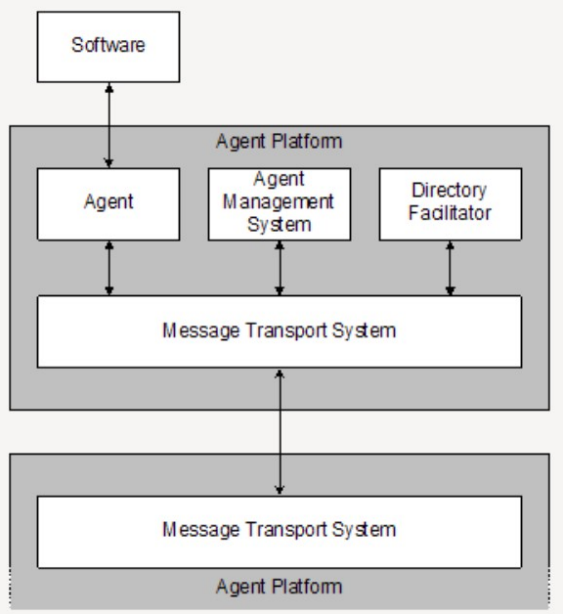
I. A. Ferguson, "Touring Machines: autonomous agents with attitudes," in Computer, vol. 25, no. 5, pp. 51-55, (1992)
Müller, J. and Markus Pischel. “The agent architecture InteRRaP: concept and application.” (1993).





FIPA platform for MAS

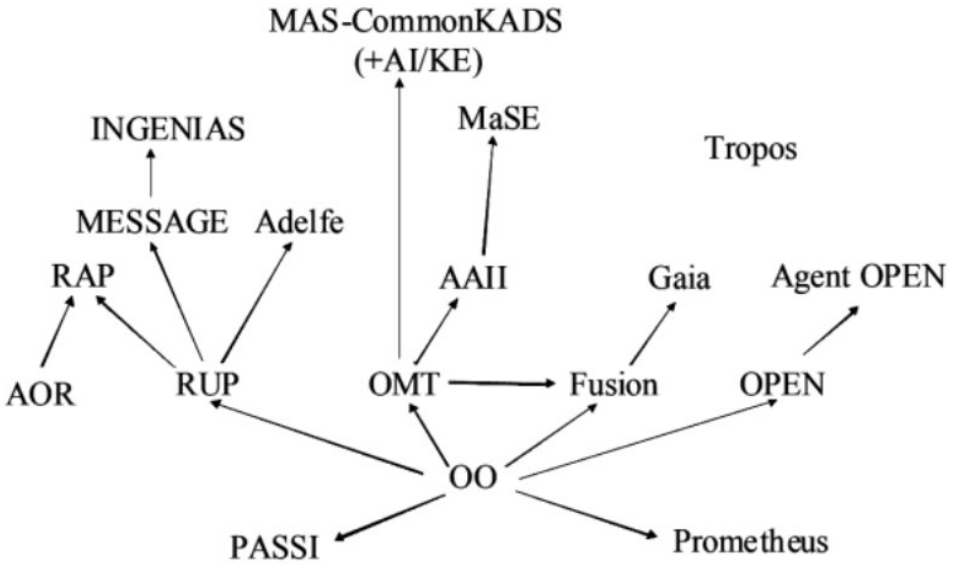
FIPA was created in Swiss in 1996 and accepted as IEEE standards committee 2005

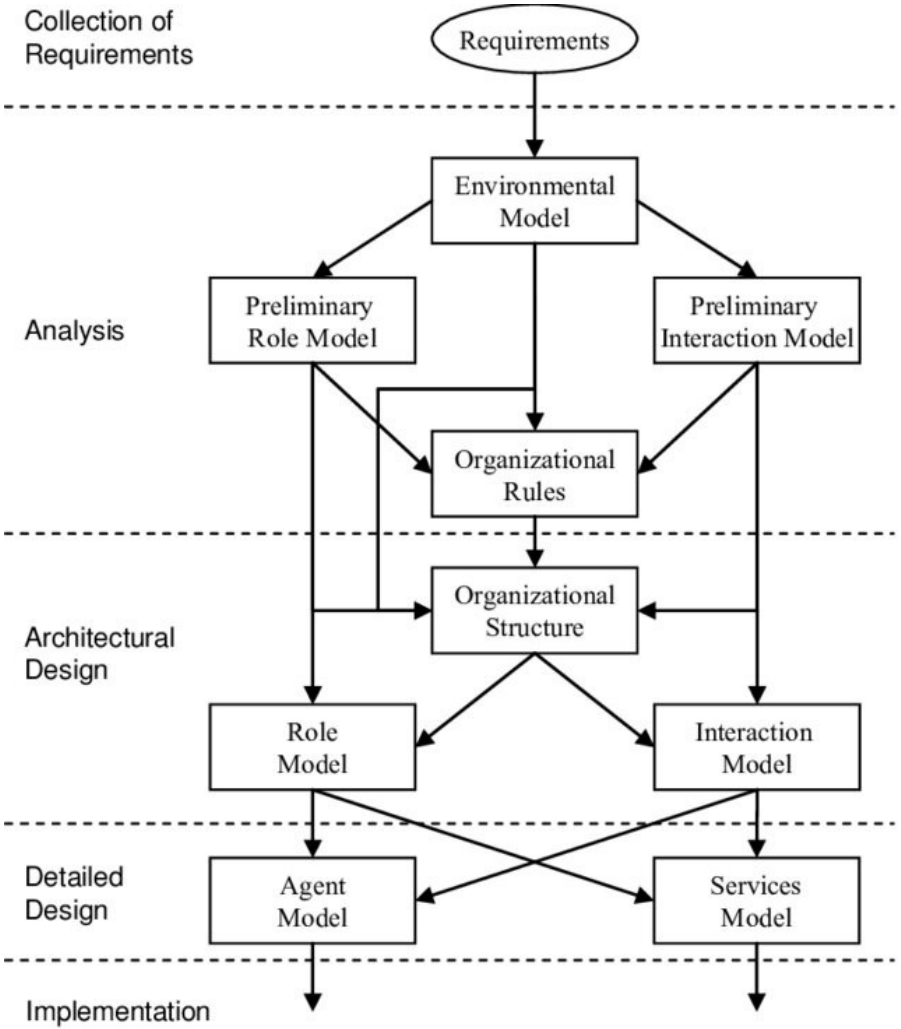


AOSE methodologies and notation techniques

AOSE methodologies

Henderson-Sellers, Brian, and Paolo Giorgini, eds. Agent-oriented methodologies. Igi Global, 2005.



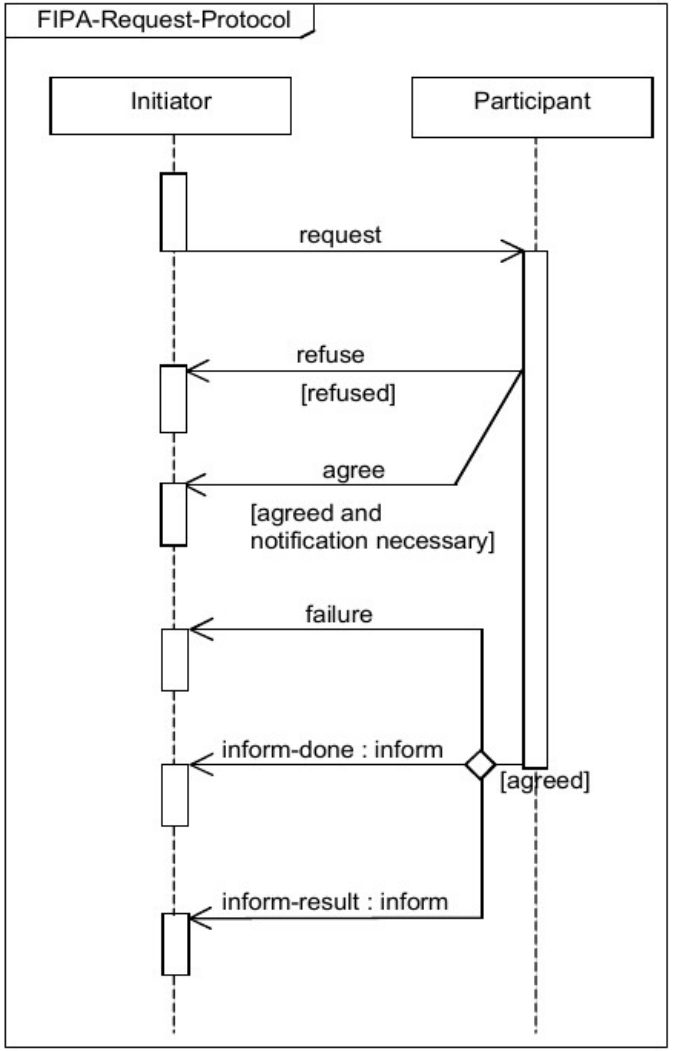


Gaia

Wooldridge, M., Jennings, N.R. & Kinny, D.
The Gaia Methodology for Agent-Oriented Analysis and Design. Autonomous Agents and Multi-Agent Systems 3, 285–312 (2000).

AOSE notations

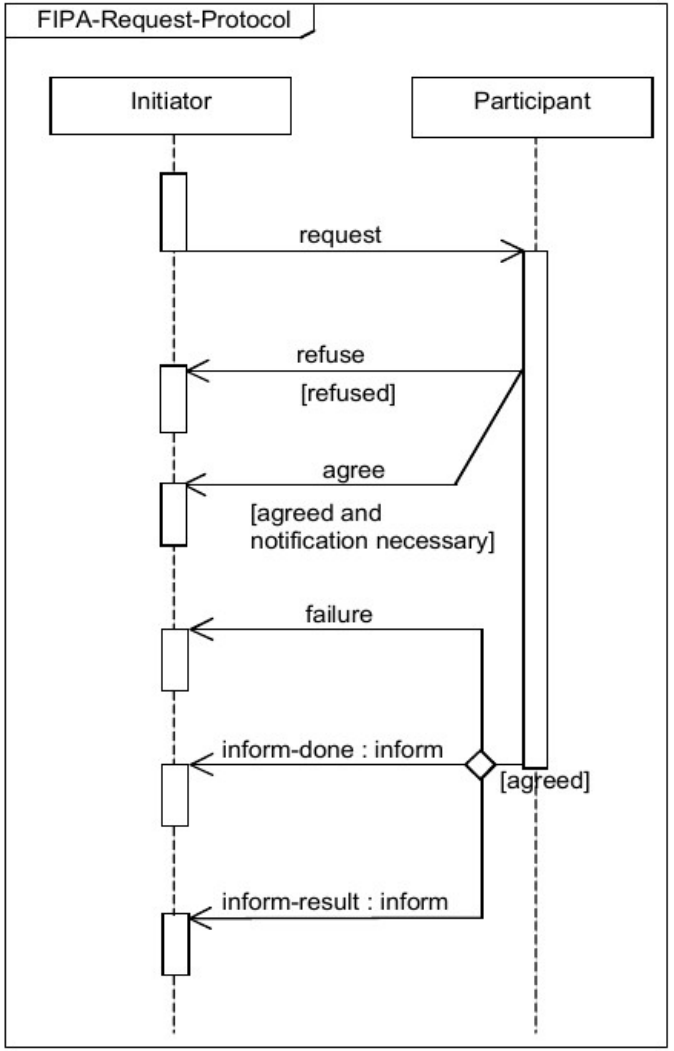
Odell J, Parunak HV, Bauer B. Extending UML for agents. 2000



AOSE notations

Odell J, Parunak HV, Bauer B. Extending UML for agents. 2000

Remember this protocol!



FIPA ACL specification

Parameter	Category of Parameters
Performative	Type of communicative acts
Sender	Participant in communication
Receiver	Participant in communication
reply-to	Participant in communication
Content	Content of message
Language	Description of Content
Encoding	Description of Content

REQUEST	The sender asks the receiver to perform some action.
AGREE	The sender agrees to perform the requested action
REFUSE	The sender refuses to perform the requested action.
NOT_UNDERSTOOD	Someone participating in the conversation didn't understand something
INFORM	A positive reply because an action is performed or the answer to some pervious question
FAILURE	A failure in performing the requested action happened after an initial agreement.

AOSE notations

Prometheus, Winikoff, M., Padgham, L. The Prometheus Methodology. 2004

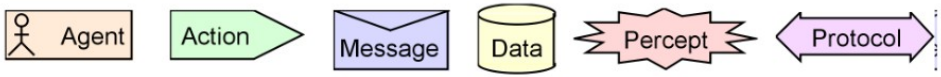


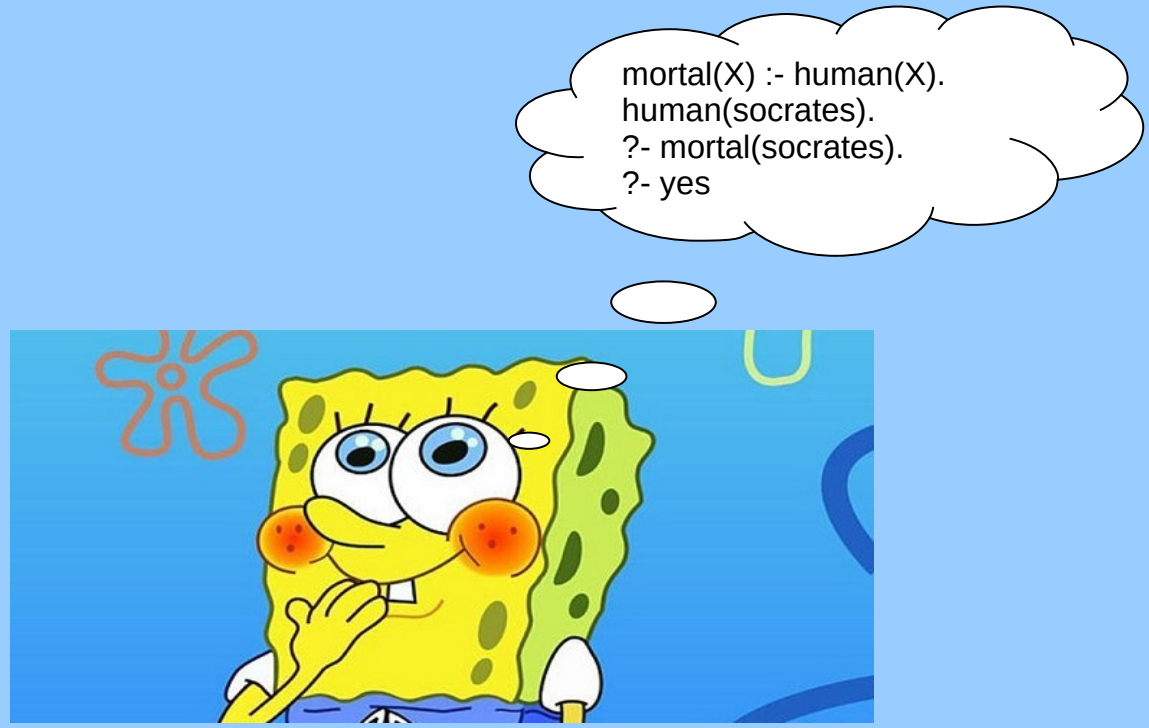
Figure 3: Notation used in System Overview Diagram



Figure 4: Notation used in Agent Overview Diagrams

Agent-oriented programming languages

Logic-based agents



Logic-based agents

“Traditional” approach to build artificial intelligent systems:

Logical formulae: symbolic representation of the agent's environment and desired behavior.

Logical deduction or **theorem proving**: syntactical manipulation of this representation.

AgentSpeak(L)

The BDI architecture is one of the best known and most studied architectures for cognitive agents.

AgentSpeak(L) is a logic-based programming language inspired by the BDI architecture.

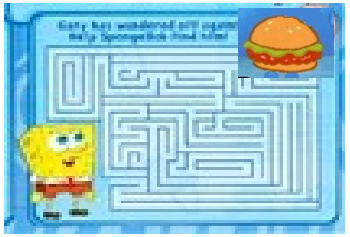
Anand S. Rao. AgentSpeak(L): BDI agents speak out in a logical computable language. In Proc. of MAAMAW '96, pp. 42-55, 1996.

Beliefs represent the knowledge of the agents.

Goals are beliefs, or conjunctions and disjunctions of beliefs, which must be achieved or tested in the current state.

Plans contain the procedural knowledge of agents. They are characterized by

- a trigger;
- a context;
- a body;
- a maintenance condition;
- a set of “success actions”;
- a set of “failure actions”.



Intentions are partially instantiated plans.

AgentSpeak(L) syntax

$$
\begin{array}{lll}
ag & ::= & bs \quad ps \\
bs & ::= & at_1 \ldots at_n \qquad (n \geq 0) \\
at & ::= & P(t_1, \ldots t_n) \qquad (n \geq 0) \\
ps & ::= & p_1 \ldots p_n \qquad (n \geq 1) \\
p & ::= & te : ct \leftarrow h \\
te & ::= & +at \mid -at \mid +g \mid -g \\
ct & ::= & at \mid \neg at \mid ct \ \wedge \ ct \mid \mathsf{T} \\
h & ::= & a \mid g \mid u \mid h; h \\
g & ::= & !at \mid \ ?at \\
u & ::= & +at \mid -at
\end{array}
$$

AgentSpeak(L) semantics

Definition 6. An *agent* is given by a tuple <E,B,P,I,A,$\mathcal{S}_\mathcal{E}$,$\mathcal{S}_\mathcal{O}$,$\mathcal{S}_\mathcal{I}$>, where E is a set of events, B is a set of base beliefs, P is a set of plans, I is a set of intentions, and A is a set of actions. The selection function $\mathcal{S}_\mathcal{E}$ selects an event from the set E; the selection function $\mathcal{S}_\mathcal{O}$ selects an option or an applicable plan (see Definition 10) from a set of applicable plans; and $\mathcal{S}_\mathcal{I}$ selects an intention from the set I.

The sets B, P, and A are as defined before and are relatively straightforward. Here we describe the sets E and I.

Definition 7. The set I is a set of intentions. Each *intention* is a stack of *partially instantiated plans*, i.e., plans where some of the variables have been instantiated. An intention is denoted by $[p_1 \ddagger \ldots \ddagger p_z]$, where p_1 is the bottom of the stack and p_z is the top of the stack. The elements of the stack are delimited by ‡. For convenience, we shall refer to the intention `[+!true:true <- true]` as the *true intention* and denote it by T.

Definition 8. The set E consists of events. Each event is a tuple $<e, i>$, where e is a triggering event and i is an intention. If the intention i is the *true intention*, the event is called an *external event*; otherwise it is an *internal event*.

Now we can formally define the notion of relevant and applicable plans and unifiers. As we saw earlier, a triggering event d from the set of events, E, is to be unified with the triggering event of all the plans in the set P. The *most general unifier (mgu)* that unifies these two events is called the relevant unifier. The intention i could be wither the true intention or an existing intention which triggered this event. More formally,

[3] The reader can refer to the Appendix for some basic definitions from first-order logic and horn clause logic.

JadeScript

The agent-oriented programming language for JADE agents.

Federico Bergenti, Giovanni Caire, Stefania Monica, Agostino Poggi. The first twenty years of agent-based software development with JADE. Autonomous Agents and Multi-Agent Systems, 34(2):36, 2020.

https://github.com/aiagents/jadescript

JadeScript

```
agent Ping
    uses ontology PingPong

    on create do
        log "Agent 'Ping' created."
        activate WaitFromPong
        activate SendRequest
```

```
agent Pong
    uses ontology PingPong

    on create do
        log "Agent 'Pong' created."
        activate WaitAndReply
```

```
ontology PingPong
    action Reply
    proposition Received
```

JadeScript

```
cyclic behavior WaitAndReply
    for agent Pong

    on message request Reply do
        sender = sender of message
        log "received 'Reply' request from agent '"
            + name of sender + "'"
        inform Received to sender


one shot behavior SendRequest
    for agent Ping

    on execute do
        log "sending 'Reply' request..."
        send message request Reply to "Pong"@
```

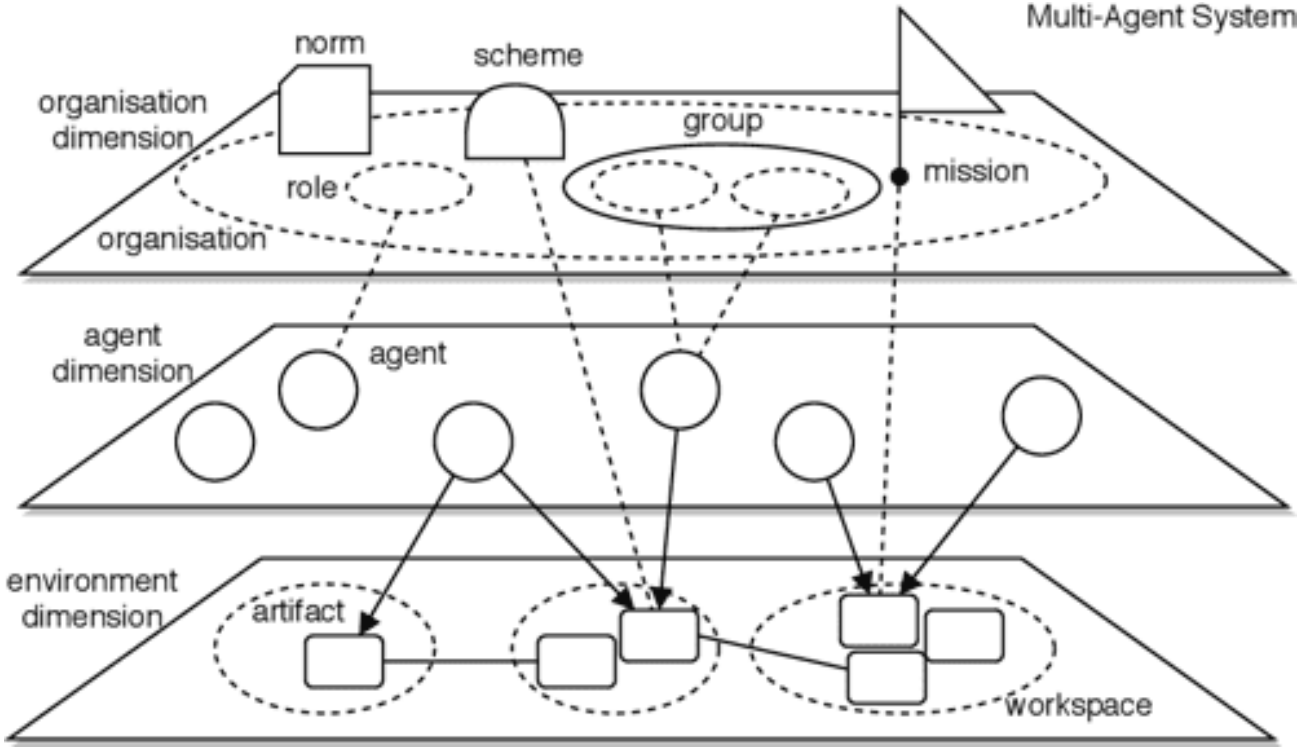
MAS infrastructures and toolkits

Wednesday 28th August @ EUMAS

Agent Toolkits Special Track

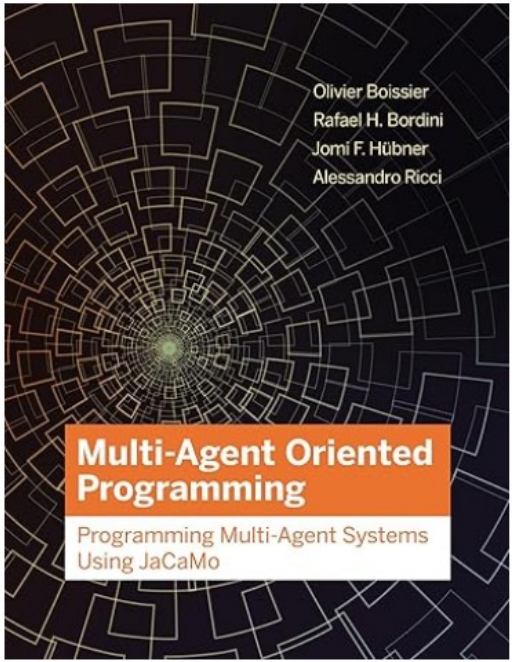
- VEsNA: From Virtual Environments via Natural language Agents to the Metaverse and Beyond: Angelo Ferrando, Andrea Gatti and Viviana Mascardi
- Belief Revision: The Good, The Bad, and JavaScript: Timotheus Kampik
- A Software Suite for Implementing Multiagent Systems Based on Protocols: Amit Chopra, Samuel Christie and Munindar P. Singh
- Yggdrasil: An Artifact-based Framework for Hypermedia Multi-Agent Systems: Andrei Ciortea, Matteo Castellucci, Kai Schultz, Jérémy Lemée, Danai Vachtsevanou, Fabien Gandon, Simon Mayer, Valentin Berger, Samuele Burattini, Alessandro Ricci and Olivier Boissier
- Ten Years of SARL: What's Next? Sebastian Rodriguez, Stéphane Galland and Nicolas Gaud
- Feasibility Study on Using Agent Behaviours for Event-Driven Programming in Jadescript: Federico Bergenti and Stefania Monica
- Better Tools for a Mainstream BDI through JaKta: Martina Baiardi, Samuele Burattini, Giovanni Ciatto and Danilo Pianini
- Jason+BSPL: Including Communication Protocols in Jason: Matteo Baldoni, Samuel H. Christie V, Amit Chopra and Munindar P. Singh
- Exceptions and Accountability for Robust Applications of JaCaMo: Matteo Baldoni, Cristina Baroglio, Roberto Micalizio and Stefano Tedeschi
- Agents Are All You Need: Platformless Programming with Jadex V: Lars Braubach, Kai Jander and Alexander Pokahr
- Language Level Support for Learning and Explaining in BDI Agents: Katharine Beaumont and Rem Collier

JaCaMo

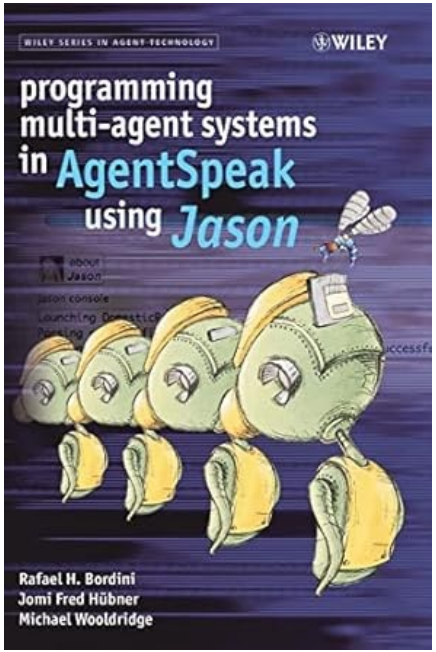


https://jacamo-lang.github.io/

JaCaMo



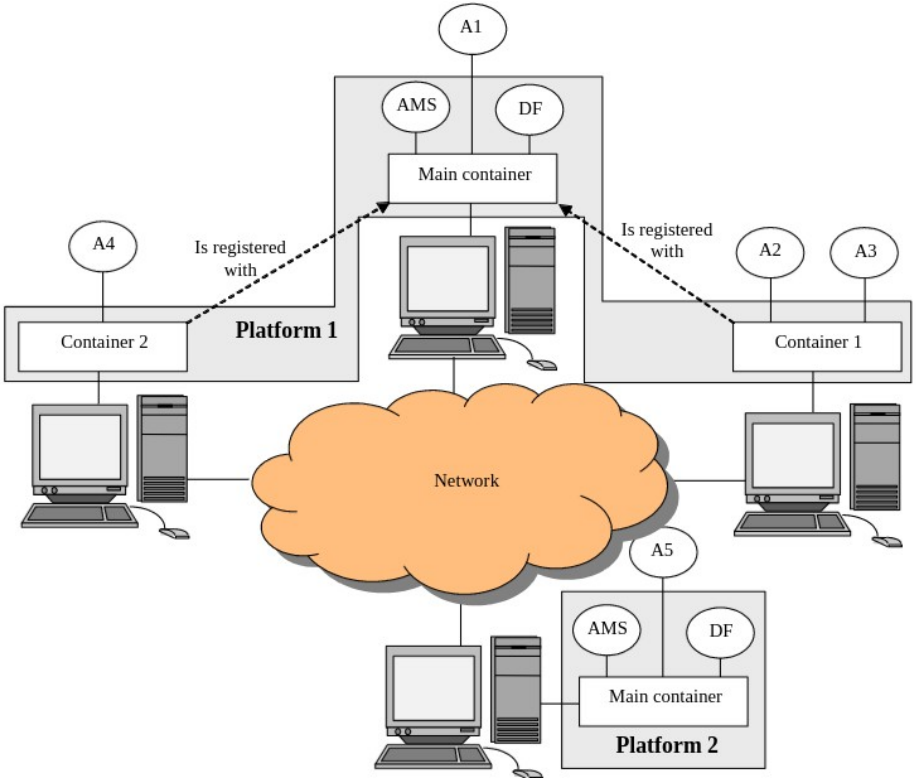
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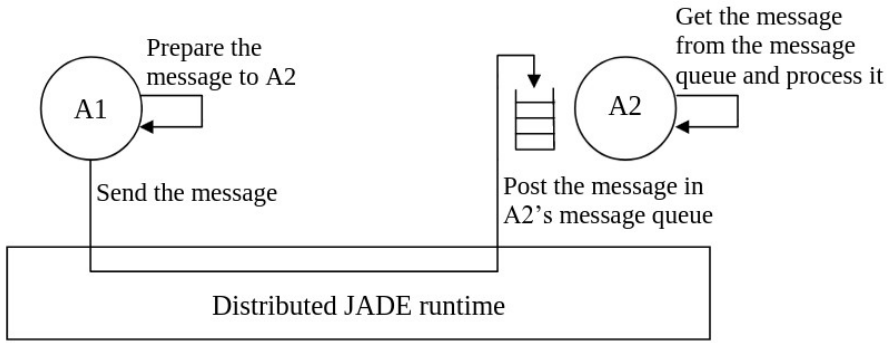


2007

https://jacamo-lang.github.io/

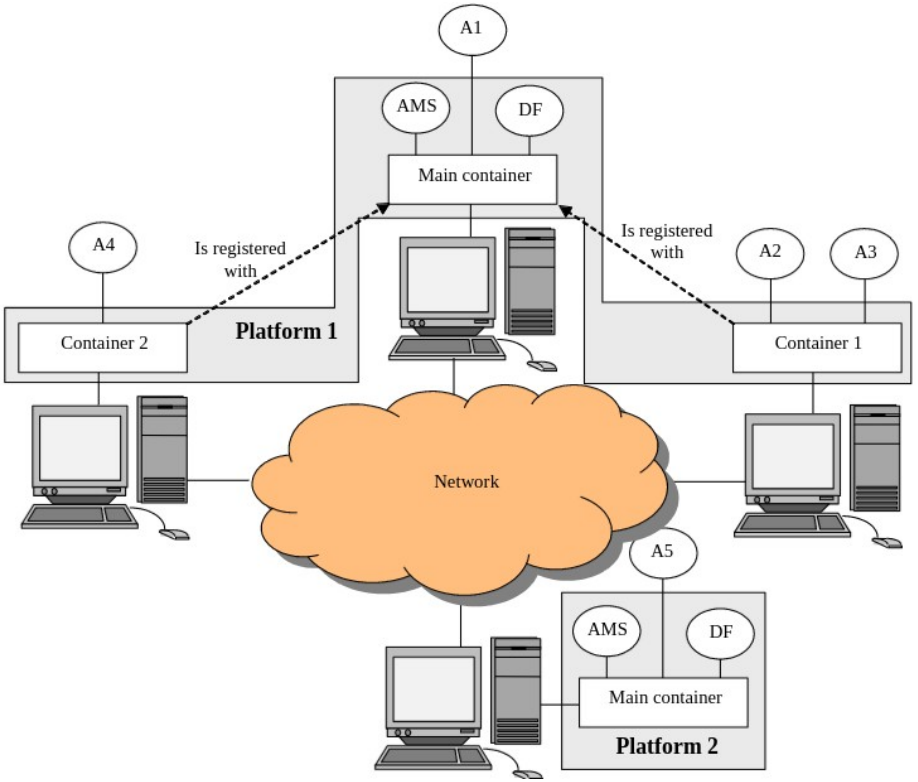
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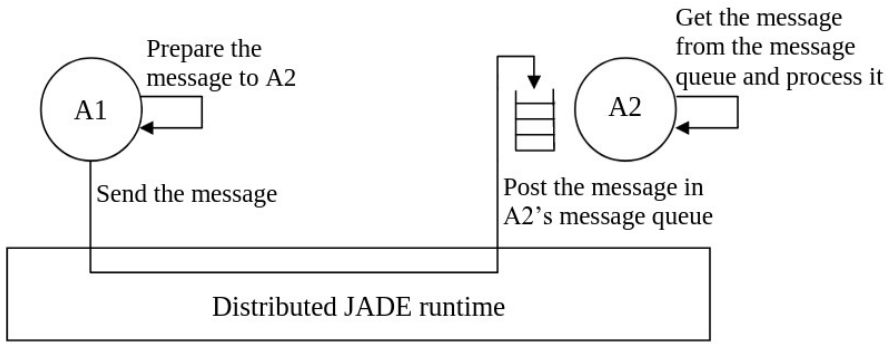


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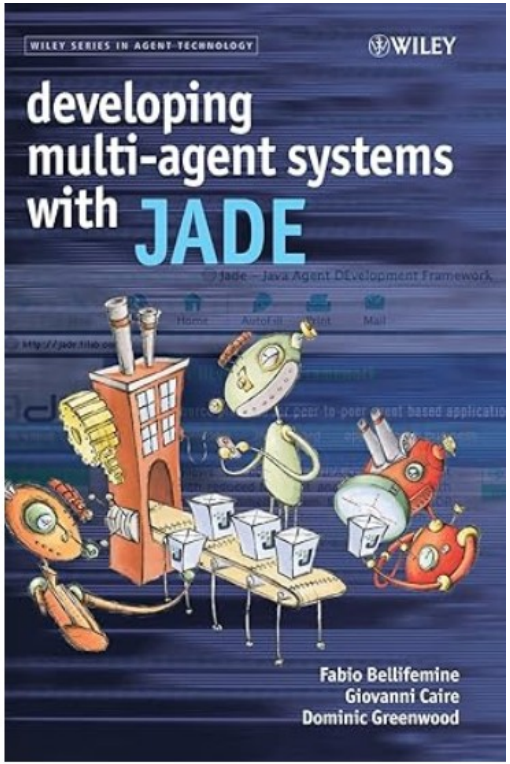


Remember this framework!



https://jade.tilab.com/

Jade



2007

https://jade.tilab.com/

Agent-based modeling and simulation

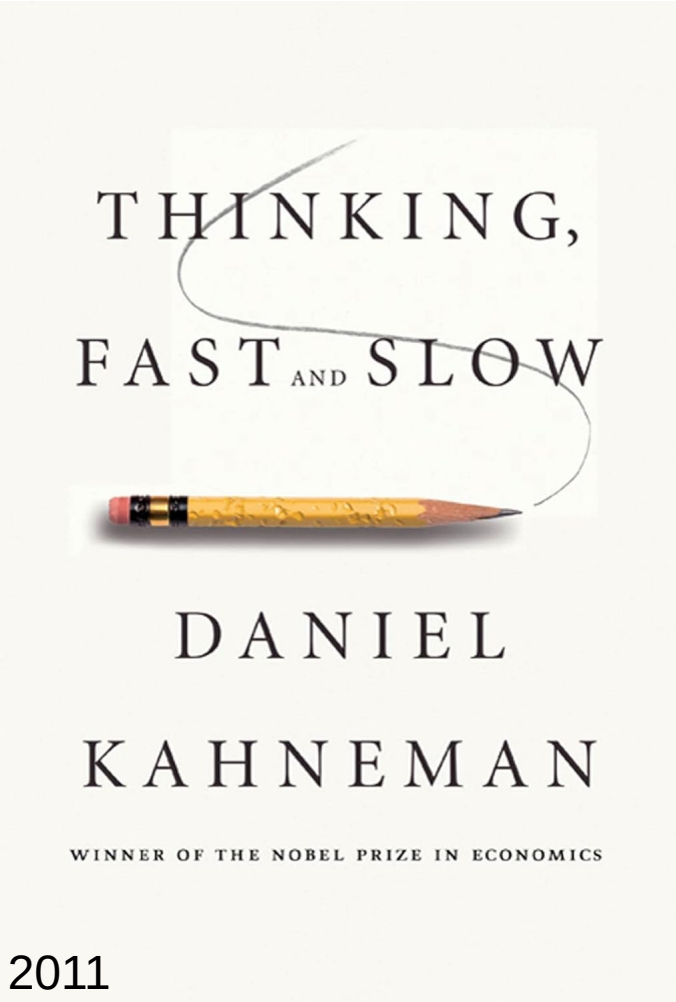
Tomorrow @ EASSS

Tutorial 6: High-Performance Computing with Agent-Based Models

Abstract: This course will briefly introduce the steps you need to take to use your agent-based model on high-performance computing facilities. It focuses on agent-based models developed using NetLogo, and potentially other environments based on the Java Virtual Machine, but contains material relevant to any agent-based model in whatever language. It comprises a cut-down version of an earlier two-day tutorial on the same subject

Tutor: Gary Polhill

Integration of symbolic and sub-symbolic approaches for agents


THINKING,
FAST AND SLOW
DANIEL
KAHNEMAN
WINNER OF THE NOBEL PRIZE IN ECONOMICS

2011

System 1: Fast, automatic, frequent, emotional, stereotypic, unconscious

Examples (in order of complexity) of things system 1 can do:

- determine that an object is at a greater distance than another
- localize the source of a specific sound
- complete the phrase "war and ..."
- display disgust when seeing a gruesome image
- solve 2+2=?
- read text on a billboard
- drive a car on an empty road
- think of a good chess move (if you're a chess master)
- understand simple sentences

Source: Wikipedia

System 2: Slow, effortful, infrequent, logical, calculating, conscious

- prepare yourself for the start of a sprint
- direct your attention towards the clowns at the circus
- look for the woman with the grey hair
- try to recognize a sound
- sustain a faster-than-normal walking rate
- determine the appropriateness of a behavior in a social setting
- count the number of A's in a certain text
- give someone your telephone number
- park into a tight parking space
- determine the price/quality ratio of two washing machines
- determine the validity of a complex logical reasoning

Source: Wikipedia

System 1 and System 2 in AI

It is possible to draw a very loose parallel between the capabilities included in system 1 and system 2, and the two main lines of work in AI: machine learning (ML) and symbolic logic reasoning, or data-driven vs knowledge-driven AI. System 1 seems to have similar traits as ML, with its ability to build (possibly imprecise and biased) models from sensory data. For example, perception activities, such as seeing and reading, that are currently addressed with ML techniques, are usually handled by human's system 1. However, system 1 exploits its ability to grasp basic notions of causality to build such models, and this does not seem to be present, at least for now, in ML. Another system 1 trait that is lacking in ML is common-sense reasoning (that of course is also related to causality). On the other hand, system 2's capability to solve complex problems seems to be related to AI techniques based on logic, search, optimization, and planning, and employing explicit knowledge, symbols, and high-level concepts.

However, as noted earlier, these two broad sets of capabilities are usually symbiotic, which supports hybrid neuro-symbolic approaches in AI.

Grady Booch, Francesco Fabiano, Lior Horesh, Kiran Kate, Jonathan Lenchner, Nick Linck, Andrea Loreggia, Keerthiram Murugesan, Nicholas Mattei, Francesca Rossi, Biplav Srivastava: Thinking Fast and Slow in AI. AAAI 2021: 15042-15046

Google Scholar

symbolic sub-symbolic integration

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[PDF] A Framework for **Integrating Symbolic** and **Sub-Symbolic** Representations.
K Clark, B Hengst, M Pagnucco, D Rajaratnam… - IJCAI, 2016 - academia.edu
… This paper establishes a framework that hierarchically **integrates symbolic** and **sub-symbolic** representations in an architecture for cognitive robotics. It is formalised abstractly as nodes …
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On the **integration** of **symbolic** and **sub-symbolic** techniques for XAI: A survey
R Calegari, G Ciatto, A Omicini - Intelligenza Artificiale, 2020 - content.iospress.com
… This is why **symbolic** approaches are getting more and more … techniques **integrating symbolic** and **sub-symbolic** approaches is … of the main **symbolic**/**sub-symbolic integration** techniques, …
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[PDF] **Symbolic** vs **sub-symbolic** ai methods: Friends or enemies?
E Ilkou, M Koutraki - CIKM (Workshops), 2020 - researchgate.net
… the approaches that combine both **symbolic** and **sub-symbolic** methods, namely, the in-… **symbolic** and **sub-symbolic** techniques which run in parallel, while the **integrated** neural **symbolic** …
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Symbolic and **sub-symbolic** representations in computational models of human cognition: what can be learned from biology?
TD Kelley - Theory & Psychology, 2003 - journals.sagepub.com
… This paper argues for an **integrated symbolic** and **sub-symbolic** approach to the computational representation of cognition. It presents three arguments as evidence to bolster the idea of …
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Integrating symbolic and **sub-symbolic** reasoning
C Strannegård, AR Nizamani - … Conference, AGI 2016, New York, NY, … 2016 - Springer

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neuro-symbolic integration

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Neuro-symbolic artificial intelligence

MK Sarker, L Zhou, A Eberhart, P Hitzler - AI Communications, 2021 - content.iospress.com

… the heart of the development of **neuro-symbolic** approaches. **Neuro-Symbolic** AI has a long-… tack as the 2005 survey: this is mostly for delineation, ie we focus on **integrated** approaches. …

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One possibility of a **neuro-symbolic integration**

AV Samsonovich - Biologically Inspired Cognitive Architectures Meeting, 2021 - Springer

… Top-level logic model of the proposed **neuro-symbolic integration**. Vertical columns (pairs of boxes) in the diagram represent, respectively, 1: BICA, 2: DL, 3: EA. For description of …

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[BOOK] Perspectives of neural-symbolic **integration**

B Hammer, P Hitzler - 2007 - books.google.com

… **Neuro-symbolic integration** centers at the border of these two paradigms and tries to … Different degrees of **neuro-symbolic integration** exist:(1) Researchers incorporate aspects of …

Save Cite Cited by 123 Related articles All 7 versions

[PDF] **Neuro-Symbolic**= Neural+ Logical+ Probabilistic.

L De Raedt, R Manhaeve, S Dumancic… - NeSy …, 2019 - ugentt2k.github.io

… **neuro-symbolic** computation is to **integrate** high-level reasoning with low-level perception. We argue 1) that **neuro-symbolic** computation should **integrate** … that **neuro-symbolic integrated** …

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Neuro-symbolic approaches in artificial intelligence

P Hitzler, A Eberhart, M Ebrahimi… - National Science …, 2022 - academic.oup.com

… **Neuro-symbolic** artificial intelligence can be defined as the … Mostly, **neuro-symbolic** AI utilizes formal logic as studied in … to motivate the study of **neuro-symbolic integration**. The first one

Machine Learning, 14, 321–331 (1994)
© 1994 Kluwer Academic Publishers, Boston. Manufactured in The Netherlands.

Extended Abstract

Combining Symbolic and Neural Learning

JUDE W. SHAVLIK SHAVLIK@CS.WISC.EDU
Computer Sciences Department, University of Wisconsin, Madison, Wisconsin, USA

Editor: D. Sleeman

Date Submitted: 17 November 1992

1. Introduction

The last ten or so years have produced an explosion in the amount of research on machine learning. This rapid growth has occurred, largely independently, in both the symbolic and connectionist (neural network) machine learning communities. Fortunately, over the last few years these two communities have become less separate, and there has been an increasing amount of research that can be considered a hybrid of the two approaches. This extended abstract reviews some of the research that combines the symbolic and neural network approaches to artificial intelligence.

We will not attempt to define precisely the essential differences between the symbolic and connectionist approaches, as that would lead to a lengthy debate far beyond the scope of this report. If some distinction is needed, we can make the coarse approximation that

Forth into SteaMas future

st(r)eam of consciousness time....



5 minutes

Requirement #1: **SteaMas citizens and tourists respect well defined civil/ethical norms**

How to meet this requirement?

We need norms, we need regulators to verify that the norms (expressed in some formalism) are respected by citizens/tourists; a way to communicate these norms to regulators, but mainly to "end users" (natural language, pictures), a way to observe what people do ("instrumentation" of the system to monitor)

5 minutes



Requirement #2: **no private traffic exists: citizens and tourists may take advantage of different kinds of free and effective public transportation means**

Collect feedback from people (NLU/NLG comes into play) “offline”; Google Maps integration; economical aspects: even if for free, transportation has a cost: agents should behave like in a ***market (rational)****, to negotiate how to best spend their allocated budget; prediction & prioritization; cope with* ***emotions****, preferences (in the “not rational” sense);* ***transparency*** *(how we predict traffic, what we can expect)*



5 minutes

Requirement #3: **both citizens and tourists may be driven around by a personal drone provided by the SteaMas municipality**

See what we said for traffic: emotions, geolocalization, etc

5 minutes

Can we “sell” the agent technology to Dr. Joyce?





Conclusions





A Letter from the Editor

Where Are All the Intelligent Agents?

Editor: James Hendler
Rensselaer Polytechnic Institute
hendler@cs.rpi.edu

Intelligent Readers,

In the late 1990s, many of us believed that the large-scale deployment of "intelligent agent"-based computing was right around the corner. In the early 2000s, much US and international research funding focused on making this happen. This magazine and many others had hugely popular special issues on agents, and academic conferences on agent-based computing were abuzz, with many of us "old timers" beginning to believe that the time had really come. But now, looking at what's hot on the Web, in IT development, and in venture capital circles, I find myself shaking my head and wondering, "Where are all the agents?"

IEEE INTELLIGENT SYSTEMS, 2007



Expert Opinion

The Agents Are All Busy Doing Stuff!

Peter McBurney, *University of Liverpool*
Michael Luck, *King's College London*

Jim Hendler asks, "Where have all the agents gone?" (see A Letter from the Editor in the May/June 2007 issue). The short answer to the question is: The agents are all busy doing stuff! You're just not looking for them in the right places!

It seems that the question arises because most of us, as consumers, can only readily imagine business-to-consumer applications of new technologies. Yet, the most compelling applications of agent technologies were always, are now, and will always be in business-to-business (B2B) domains, out-of-sight of most consumers.

The latest weekly statistics from the New York Stock Ex-

the agents?" is to misunderstand these technologies' nature and scope.

Asking this question also risks misunderstanding the pervasiveness of these technologies' use and deployment. Many companies founded their business on agent technologies: what they sell is not just a conceptualization but a conceptualization allied to specific agent-related technologies. So Eurobios, for example, applies optimization technologies that can be used in the context of the conceptualization of agents. Agentis Software offers goal-directed technologies based on the BDI (belief-desire-intention) model, as does the Agent Oriented Software Group. Similarly, Lost Wax, Magenta Technology, and Whitestein Technologies (and Living Systems) have technology platforms or products that

IEEE INTELLIGENT SYSTEMS, 2007



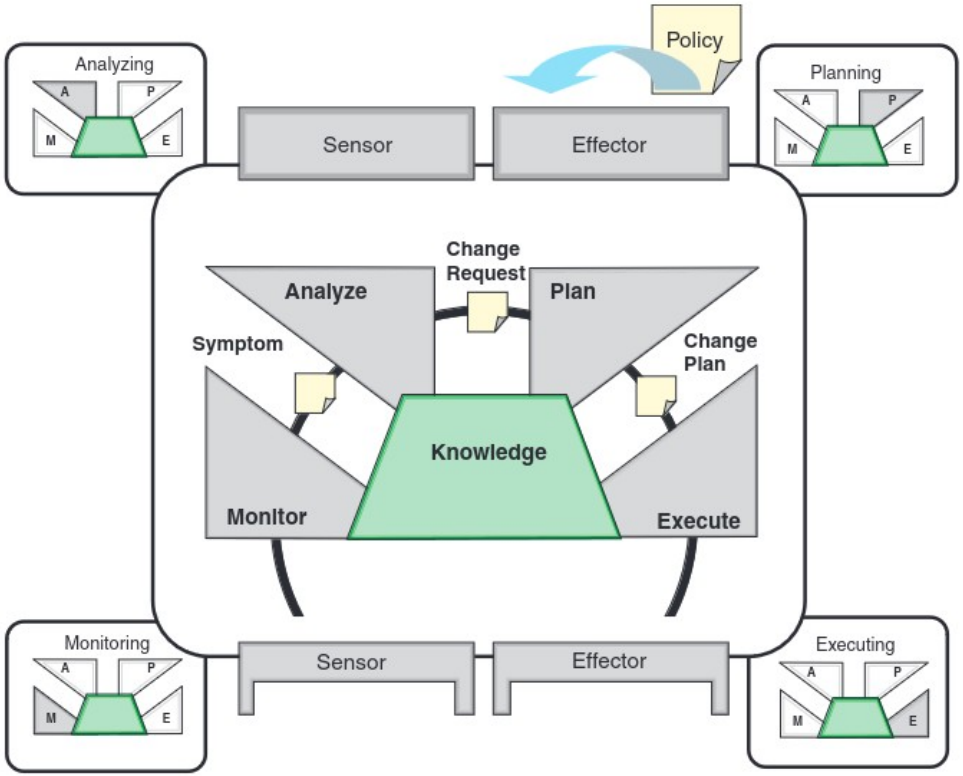


Figure 4. Functional details of the autonomic manager

touchpoints, provide sensor and effector manageability interfaces for other autonomic managers and manual managers to use. Using standard sensor and effector interfaces enables these components to be composed together in a manner that is transparent to the managed resources. For example,



Autonomic Computing
White Paper

An architectural blueprint for autonomic computing.

Third edition
June 2005



Agents **are** all around: sometimes presented with different names, often deployed using mainstream technologies and languages (it's ok: recall the Gaia methodology that leaves the implementation stage free)

If you will take agents and MAS as a "mindset", you will find them everywhere, and you will understand the huge potential for their use



ELSEVIER

Computer Methods and Programs in Biomedicine

Volume 236, June 2023, 107525

A survey on agents applications in healthcare: Opportunities, challenges and trends

Emilio Sulis [a], Stefano Mariani [b], Sara Montagna [c]

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Deep Reinforcement Learning for Multiagent Systems: A Review of Challenges, Solutions, and Applications

Thanh Thi Nguyen, Ngoc Duy Nguyen, and Saeid Nahavandi, *Senior Member, IEEE*

***Abstract*—Reinforcement learning (RL) algorithms have been around for decades and employed to solve various sequential decision-making problems. These algorithms, however, have faced great challenges when dealing with high-dimensional environments. The recent development of deep learning has enabled RL methods to drive optimal policies for sophisticated and capable agents, which can perform efficiently in these challenging environments. This article addresses an important aspect of deep RL related to situations that require multiple agents to communicate and cooperate to solve complex tasks. A survey of different approaches to problems related to multiagent deep RL (MADRL) is presented, including nonstationarity, partial observability, continuous state and action spaces, multiagent training schemes, and multiagent transfer learning. The merits and demerits of the reviewed methods will be analyzed and discussed with their corresponding applications explored. It is envisaged that this review provides insights about various MADRL methods and can lead to the future development of more robust and highly useful multiagent learning methods for solving real-world problems.**

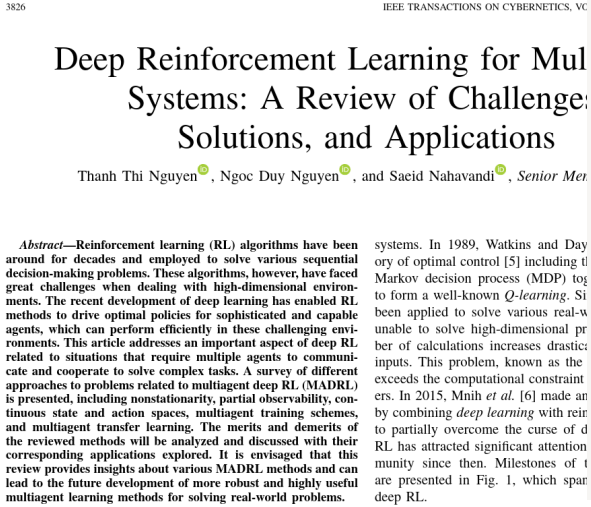
systems. In 1989, Watkins and Dayan [4] brought the theory of optimal control [5] including the *Bellman equation* and Markov decision process (MDP) together with TD learning to form a well-known *Q-learning*. Since then, *Q*-learning has been applied to solve various real-world problems, but it is unable to solve high-dimensional problems where the number of calculations increases drastically with the number of inputs. This problem, known as the *curse of dimensionality*, exceeds the computational constraint of conventional computers. In 2015, Mnih *et al.* [6] made an important breakthrough by combining *deep learning* with reinforcement learning (RL) to partially overcome the curse of dimensionality. The deep RL has attracted significant attention from the research community since then. Milestones of the development of RL are presented in Fig. 1, which span the TE method to the deep RL.



Agents **are** all around: sometimes presented with different names, often deployed using mainstream technologies and languages (it's ok: recall the Gaia methodology that leaves the implementation stage free)

If you will take agents and MAS as a "mindset", you will find them everywhere, and you will understand the huge potential for their use







Thank you for your attention!

Questions?



You can find me at
viviana.mascardi@unige.it

Engineering Safe Multiagent Systems: an Introduction to Formal and Runtime Verification of Agents and MAS



Davide Ancona, University of Genova, Italy

Viviana Mascardi, University of Genova, Italy

EASSS 2024, August 22nd

Safe and reliable autonomous systems

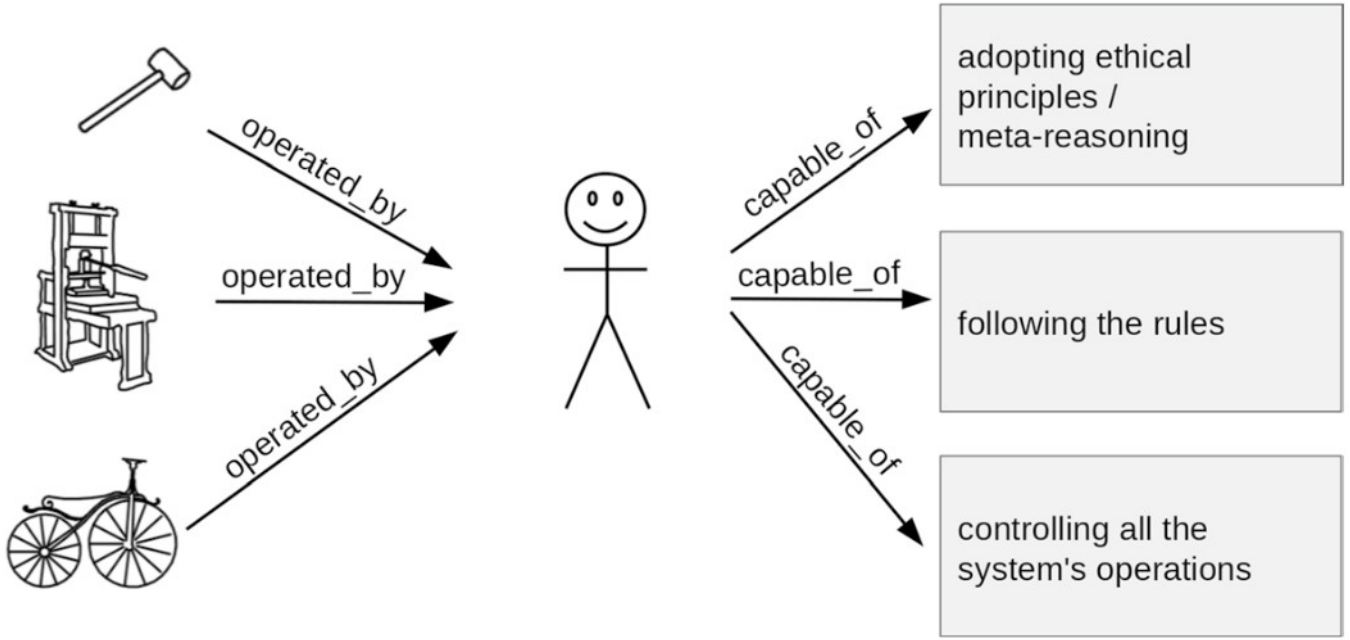


Fig. 1 Human user has the full control of the tool

[Towards a Framework for Certification of Reliable Autonomous Systems: Michael Fisher, Viviana Mascardi, Kristin Yvonne Rozier, Bernd-Holger Schlingloff, Michael Winikoff, Neil Yorke-Smith, JAAMAS 2021]

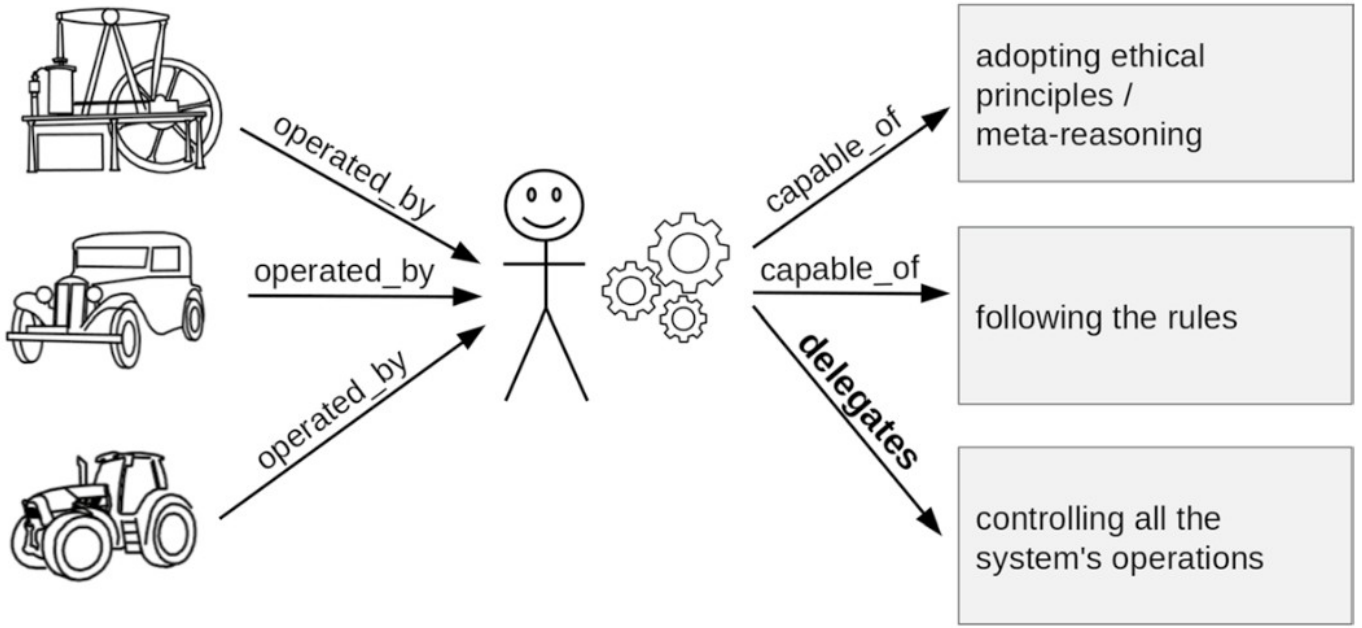


Fig. 2 Human (partially) delegates the control of the operations

[Towards a Framework for Certification of Reliable Autonomous Systems: Michael Fisher, Viviana Mascardi, Kristin Yvonne Rozier, Bernd-Holger Schlingloff, Michael Winikoff, Neil Yorke-Smith, JAAMAS 2021]

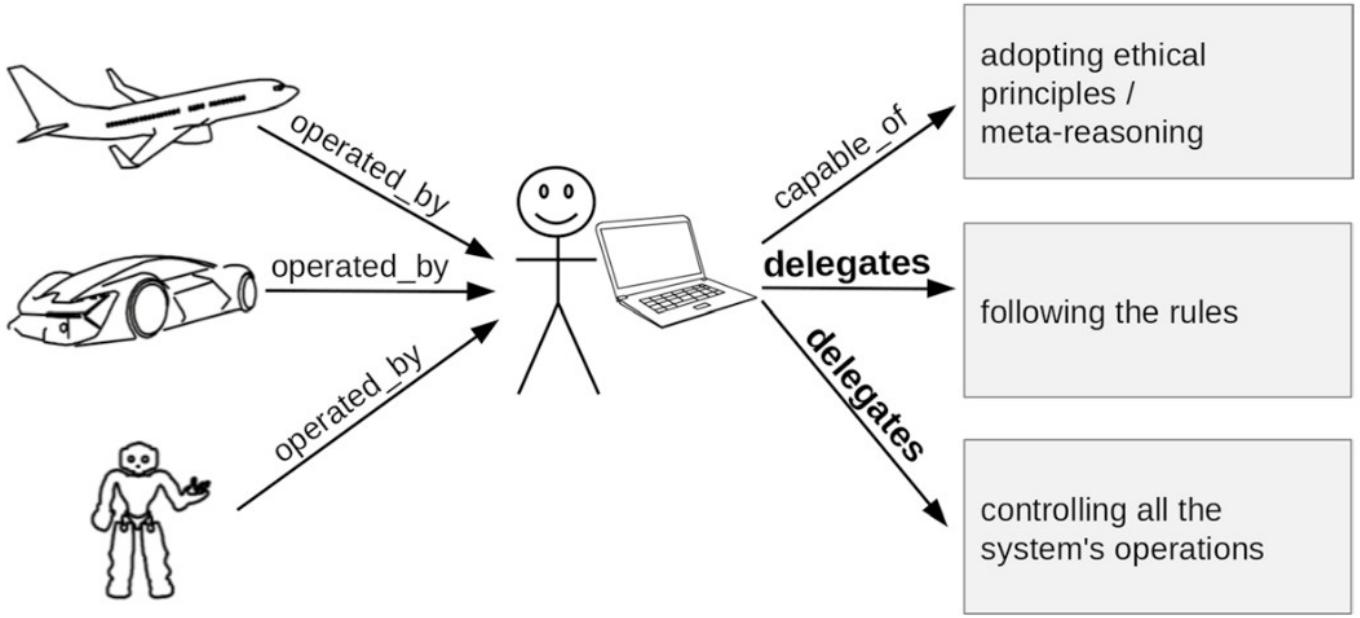


Fig. 3 Human (partially) delegates the respect of the rules

[Towards a Framework for Certification of Reliable Autonomous Systems: Michael Fisher, Viviana Mascardi, Kristin Yvonne Rozier, Bernd-Holger Schlingloff, Michael Winikoff, Neil Yorke-Smith, JAAMAS 2021]

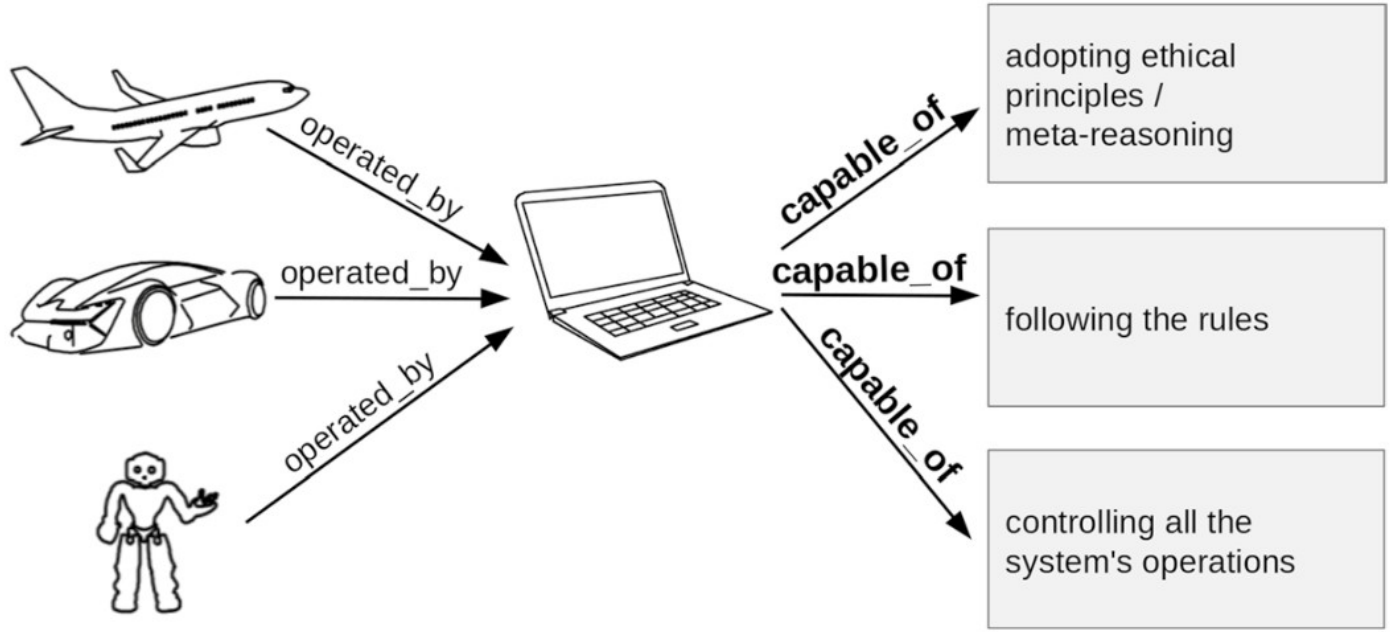


Fig. 4 Human replaced by an autonomous software system

[Towards a Framework for Certification of Reliable Autonomous Systems: Michael Fisher, Viviana Mascardi, Kristin Yvonne Rozier, Bernd-Holger Schlingloff, Michael Winikoff, Neil Yorke-Smith, JAAMAS 2021]

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	SAE LEVEL 0™	SAE LEVEL 1™	SAE LEVEL 2™	SAE LEVEL 3™	SAE LEVEL 4™	SAE LEVEL 5™
What does the human in the driver's seat have to do?	**You are driving** whenever these driver support features are engaged – even if your feet are off the pedals and you are not steering			**You are not driving** when these automated driving features are engaged – even if you are seated in "the driver's seat"		
	You must constantly supervise these support features; you must steer, brake or accelerate as needed to maintain safety			When the feature requests, you must drive	These automated driving features will not require you to take over driving	
	These are driver support features			**These are automated driving features**		
What do these features do?	These features are limited to providing warnings and momentary assistance	These features provide steering **OR** brake/ acceleration support to the driver	These features provide steering **AND** brake/ acceleration support to the driver	These features can drive the vehicle under limited conditions and will not operate unless all required conditions are met		This feature can drive the vehicle under all conditions
Example Features	• automatic emergency braking • blind spot warning • lane departure warning	• lane centering **OR** • adaptive cruise control	• lane centering **AND** • adaptive cruise control at the same time	• traffic jam chauffeur	• local driverless taxi • pedals/ steering wheel may or may not be installed	• same as level 4, but feature can drive everywhere in all conditions

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What You Need to Know About Level 3 Autonomy

You can now buy a Level 3 car in the U.S. — if you live in one of two states and can afford a new Mercedes-Benz.

David Gluckman | Jun 21, 2024 | 6 min read



Mercedes-Benz

QuickTakes:

Level 3 is a step toward full autonomy

Automated driving requires lots of sensors

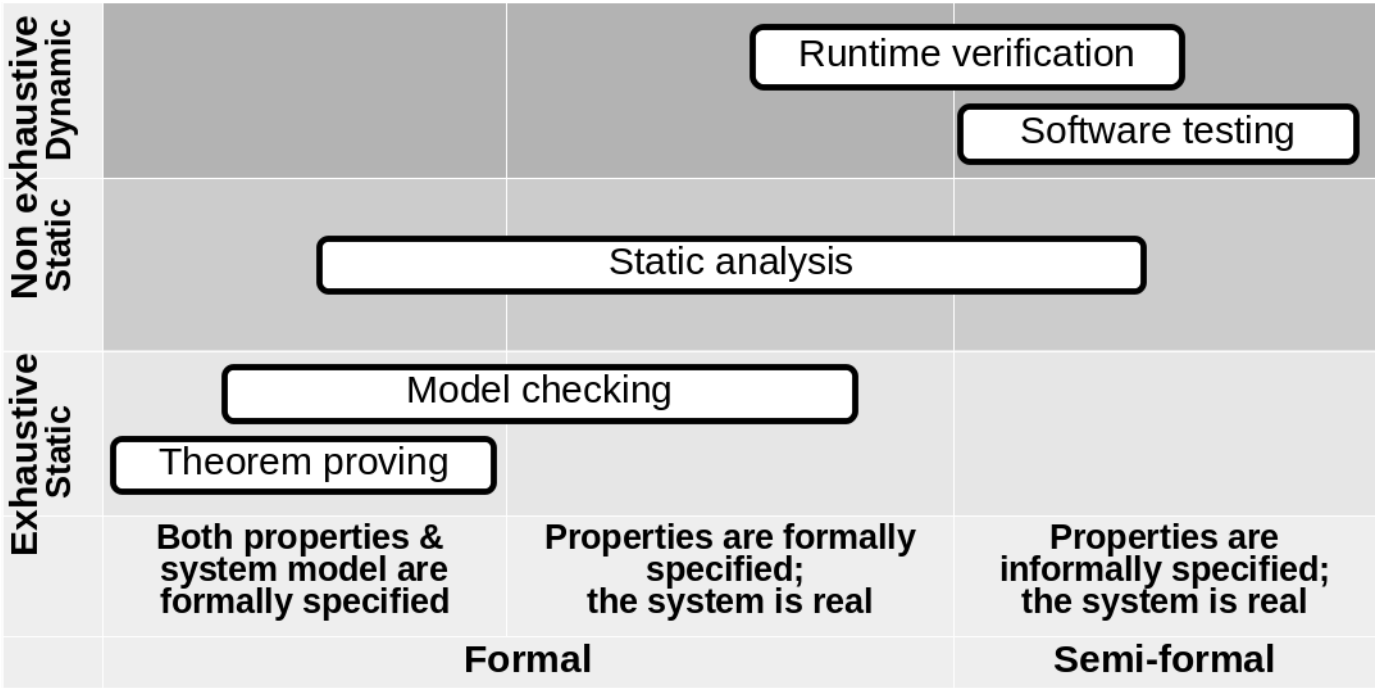
Mercedes-Benz is the only company offering Level 3 vehicles

Wide adoption of Level 3 will require regulators to catch up

Automakers have been promising fully autonomous cars for years, but only now are the technology and necessary regulations catching up. That said, not all autonomous vehicles are the same, and the first cars that let you pay attention to something other than driving — vehicles with so-called **Level 3** driving automation — have limited application. They work on only a small selection of roads, and they're sold only in certain states.



Formal & semi-formal verification techniques



[Towards a Framework for Certification of Reliable Autonomous Systems: Michael Fisher, Viviana Mascardi, Kristin Yvonne Rozier, Bernd-Holger Schlingloff, Michael Winikoff, Neil Yorke-Smith, JAAMAS 2021]

Runtime Verification

RV: some introductory reading

A Brief Account of Runtime Verification, M. Leucker, C. Schallhart, 2009

Introduction to Runtime Verification, E. Bartocci, Y. Falcone, A. Francalanza, G. Reger, 2018

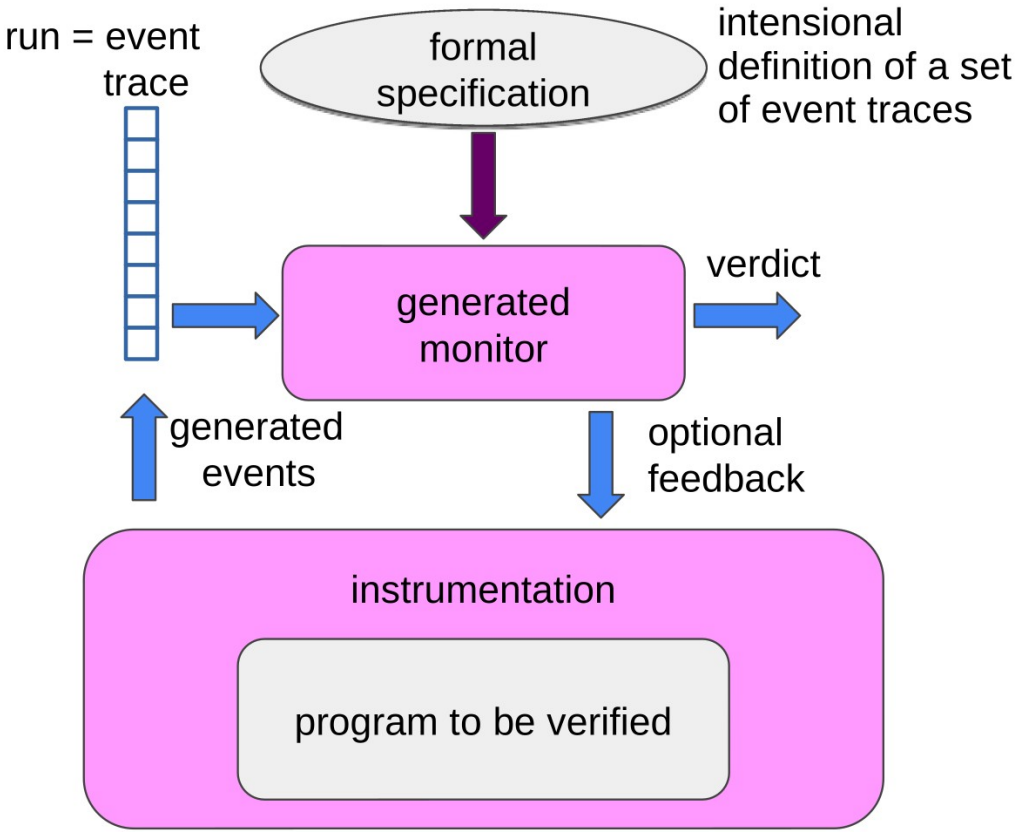
Falcone, Y., Krstić, S., Reger, G. et al. A taxonomy for classifying runtime verification tools, 2021

RV: some introductory reading



Falcone, Y., Krstić, S., Reger, G. et al. A taxonomy for classifying runtime verification tools, 2021

Runtime Verification Process



RV bridges the gap between formal verification and testing

- The notion of event trace abstracts over system executions.
- RV can be adopted online and offline.
- System agnosticism better supports reuse and interoperability.
- RV offers more opportunities for addressing error recovery, self-adaptation, and issues that go beyond software reliability.
- Often, some information is available only at runtime. In such cases, RV is an alternative to formal verification.
- The behavior of an application may depend heavily on the environment of the target system. In this scenario, RV adds on formal correctness proofs.
- In the case of systems where security is important, it is useful also to monitor behavior of properties even when they have been statically proved or tested.



The Runtime Monitoring Language

Davide Ancona, Luca Franceschini, Angelo Ferrando, Viviana Mascardi:
RML: Theory and practice of a domain specific language for runtime verification.
Sci. Comput. Program. 205: 102610 (2021)

https://rmlatdibris.github.io/

Conclusions

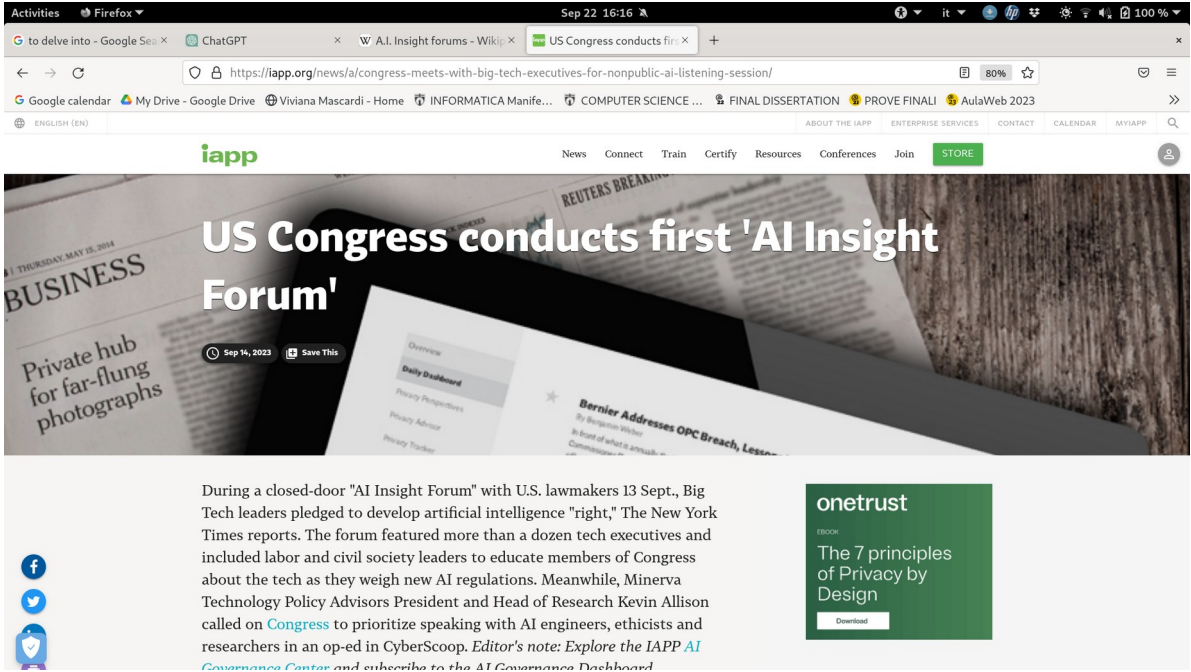


The adoption of Artificial Intelligence (and agents and MAS are AI systems) raises ethical concerns that must be coped with





The adoption of Artificial Intelligence (and agents and MAS are AI systems) raises ethical concerns that must be coped with






SAFETY AND
INDUSTRY 4.0




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SAFETY
FIRST




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INDUSTRY 4.0


SAFETY
FIRST


SAFETY FIRST



A **reliable approach for engineering applications that involve agents in safety-critical/ethical critical domains is needed**: formal methods in general, and runtime verification in particular, may help addressing some of their ethical threats, paving the way to a more explainable, trustworthy, socially responsible AI



Engineering safe agents and MAS is only one of the many goals that **AOSE** addresses, by facing the theoretical and technological challenges of nowadays complex, distributed, heterogeneous, autonomous systems

Thank you for your attention!

Questions?



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Credits: Angelo Ferrando,
University of Modena-Reggio Emilia, IT, angelo.ferrando@unimore.it